

Predicting Subscriber Churn Before It Happens

A machine-learning early-warning system for the subscriber base

71%

of churners identified

4.7%

of subscribers to contact

0.977

ROC-AUC on unseen data

Prepared by the Data Science Team

September 2026

Based on 64,738 subscribers | 3,241 churn events in a trailing 90-day window

1. Executive summary

We can now tell which subscribers are likely to leave in the next 90 days — and reach about 7 in 10 of them by contacting fewer than 5 in 100 subscribers.

5.0%

Baseline churn, 90 days
3,241 of 64,738 subscribers

0.977

ROC-AUC on unseen data
1.0 = perfect, 0.5 = coin toss

71%

of churners caught
by Tier A, 4.7% of the base

75%

of Tier A flags are real
3 in 4 outreach calls hit a churner

The challenge. Only about 1 in 20 subscribers churns in a given quarter. That makes churn hard to see: a broad retention campaign spends most of its budget on customers who were never going to leave, while the customers who do leave give little warning.

What we built. A churn-risk model that scores every subscriber using their recharge pattern, number-porting (MNP) enquiries, complaints and payment history, network experience and plan details. It was trained on 70% of the data, tuned on 15%, and judged on a final 15% that it had never seen.

What it delivers. On that unseen group of 9,711 subscribers (486 of whom churned), the model's highest-risk tier contained 460 subscribers — and 344 of the 486 churners were among them (75% of that tier actually churned, against a 5.0% average).

Find leavers early

MNP enquiries and recharge gaps carry most of the predictive power, and both can be tracked continuously.

Spend where it counts

Tier C is 88% of subscribers but under 10% of churners. Today's offers show no link to churn risk, so budget can be re-aimed.

Prove it before scaling

A 90-day pilot with a hold-out group turns our scenario estimates into a measured save rate and payback.

Our recommendation

- 1. Pilot Tier A.** Same-day, personalised retention outreach for the highest-risk subscribers, with a hold-out group to measure real savings.
- 2. Add a light-touch Tier B.** Low-cost automated nudges (recharge reminders, payment help) for the next-highest-risk band.
- 3. Keep the rest untouched.** Tier C holds 88% of subscribers but under 10% of churners — no retention spend needed.
- 4. Re-score monthly** and refine with more advanced models once the pilot confirms the approach.

2. The challenge and how we approached it

64,738

Subscribers analysed

prepaid and postpaid, 22 telecom circles

43

Fields audited

no duplicates, no negative values

5.01%

90-day churn rate

3,241 churners

1 : 19

Churn imbalance

the rare-event challenge

Churn is a **rare-event** problem. Of 64,738 subscribers, 61,497 (95.0%) stayed and 3,241 (5.0%) left within 90 days — a ratio of roughly 19 to 1.

Why headline accuracy would have been misleading

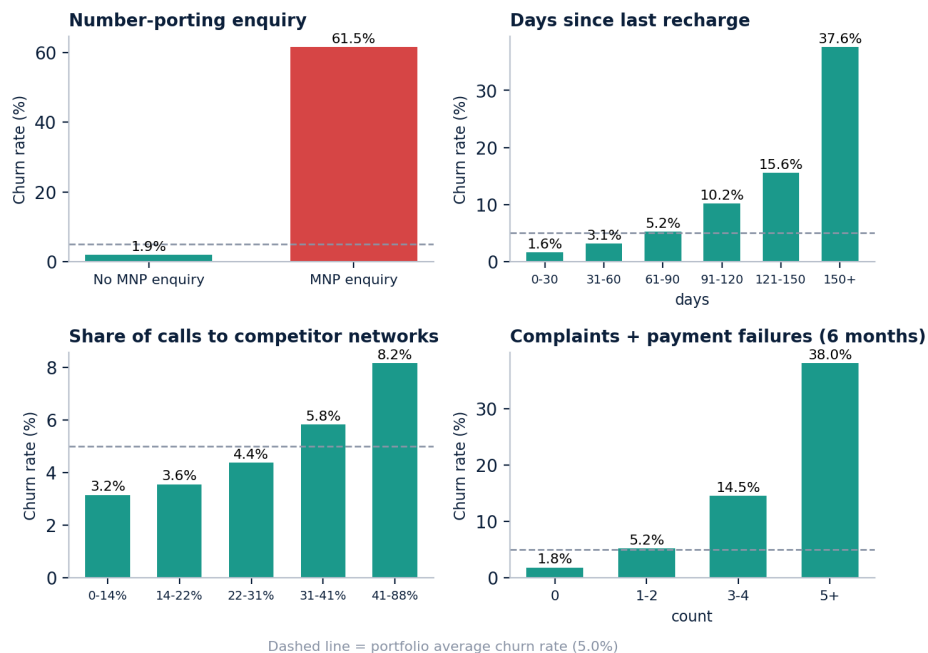
A 'model' that simply predicted *nobody will leave* would be **95.0% accurate** — and would find zero churners. We therefore judged the model on how many real churners it finds (recall), how many of its flags are correct (precision), and how well it ranks risky subscribers above safe ones (ROC-AUC and PR-AUC), rather than on accuracy alone.

How we built it

Step	What we did	Why it matters to you
1. Data audit	64,738 subscribers, 43 fields. Checked for duplicates, negative values and inconsistencies: none found.	The model learns from clean, trustworthy data.
2. Leakage protection	Removed churn_reason , churn_date and churn_flag_30d — fields that reveal the outcome after the fact. Also excluded ID and duplicate fields.	Leaving them in produced a 'perfect' 1.000 score in our exploratory test — impressive on paper, useless in real life. Excluding them keeps the results honest.
3. Fair test	Stratified 70 / 15 / 15 split: train / validation / test. Every part has the same 5.0% churn rate. The test group was used once, at the end.	Results are what you should expect on subscribers the model has never seen.
4. Transparent model	Logistic regression: a well-established, explainable baseline, with numeric fields scaled and categories encoded.	Every score can be explained to retention teams, and it sets a clear benchmark for more advanced models.
5. Imbalance handling	Class weighting so the rare churners are not ignored; decision cut-off tuned on the validation set.	The model is built to find churners, not to please an accuracy score.

3. What drives churn

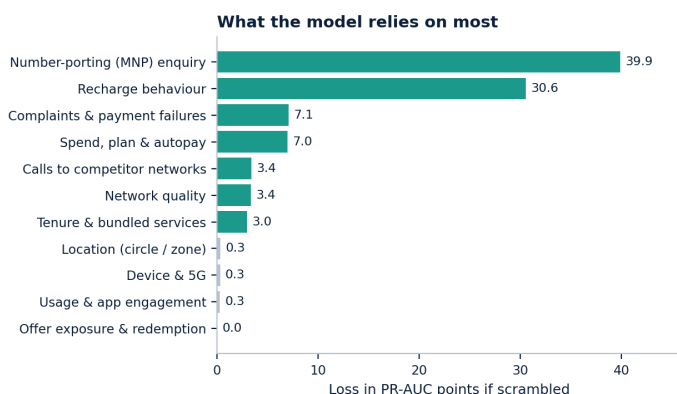
Churn is driven by behaviour and intent to leave — not by how much a subscriber spends or where they live.



Observed churn rate by subscriber behaviour across all 64,738 subscribers. Dashed line: portfolio average (5.0%).

- **Number-porting enquiry is the strongest warning.** 3,360 subscribers have an active MNP enquiry; 61.5% of them churn, versus 1.9% of everyone else — about 12 times the portfolio average.
- **Recharge gaps signal disengagement.** Churn climbs from 1.6% (recharged within 30 days) to 37.6% (150+ days since last recharge), and passes 10% once a subscriber goes beyond 90 days.
- **Friction compounds.** Subscribers with 5+ combined complaints and payment failures churn at 38%, against 1.8% for those with none.
- **Competitor-facing calling and plan type** matter too: prepaid subscribers churn at 5.6% versus 0.6% for postpaid.

How the model uses these signals



- **Two signal groups do most of the work:** MNP enquiry and recharge behaviour — also the most actionable, since both can be tracked and acted on quickly.
- **Offer exposure and redemption carry no signal** (churn is 5.02% exposed vs. 5.00% unexposed): retention offers are not currently aimed at the people most at risk — an immediate opportunity.
- **Location, device and raw usage add little** once behaviour is known; geography largely proxies for network quality.

Grouped importance on the unseen test data: how much the model's ability to find churners falls when that group of information is randomly scrambled.

4. How well does the model perform?

Measured on 9,711 subscribers the model never saw during training or tuning.

The model produces a churn-risk score for every subscriber. Where we draw the line between 'flag' and 'don't flag' is a business choice, so we show two operating modes: a **Focused** mode that prioritises accuracy of each flag, and a **Wide net** mode that prioritises catching as many churners as possible.

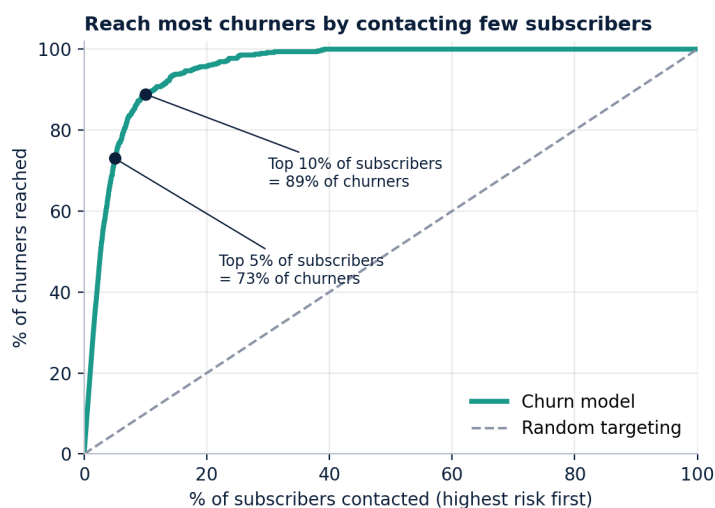
Metric	Focused mode (cut-off 0.92)	Wide-net mode (cut-off 0.50)	What it means
ROC-AUC	0.977	0.977	Ability to rank likely leavers above loyal subscribers. 1.0 = perfect; 0.5 = guessing. Same in both modes.
PR-AUC	0.809	0.809	Ranking quality on the rare churn class (random guessing scores 0.050).
Precision	74.8%	38.2%	Of the subscribers we flag, the share who genuinely would have left.
Recall	70.8%	90.7%	Of the subscribers who actually left, the share we flagged in advance.
F1-score	0.727	0.538	A single balance of precision and recall (1.0 = perfect).
Accuracy	97.3%	92.2%	Overall share of correct calls. Note the 'nobody leaves' baseline is already 95.0%.

In plain numbers (Focused mode)

	Predicted: will churn	Predicted: will stay
Actually churned (486)	344 caught	142 missed
Actually stayed (9,225)	116 false alarms	9,109 correctly left alone

Of every 100 subscribers who will leave, the model catches about 71. Of every 100 subscribers it flags, about 75 really would have left.

Reaching churners efficiently



Contact the top	Churners reached	vs. random
1% of subs	20%	20.0x
2% of subs	38%	19.2x
5% of subs	73%	14.6x
10% of subs	89%	8.9x
20% of subs	96%	4.8x

Cumulative gains on the test set: contacting subscribers in order of risk reaches churners far faster than random targeting.

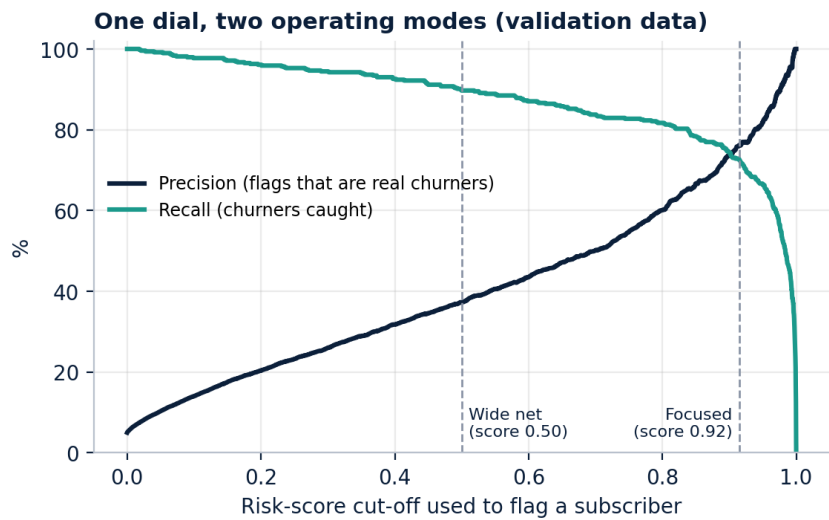
5. Putting it to work: three risk tiers

One score, three actions. The tiers translate the model into something a retention team can run every month.

Tier	Risk score	Subscribers (test group)	Share of base	Actual churn rate	Churners in tier	Share of all churners
A - Critical	0.92 or above	460	4.7%	74.8%	344	70.8%
B - Watch	0.50 to 0.92	693	7.1%	14.0%	97	20.0%
C - Stable	below 0.50	8,558	88.1%	0.5%	45	9.3%

Portfolio average churn rate: 5.0%. Tier A subscribers churn at about 15 times the average; Tier C at about one-tenth of it.

Choosing how wide to cast the net



Lowering the cut-off catches more churners but flags more subscribers who would have stayed. The tiers combine both modes: A is the Focused list, A + B is the Wide net.

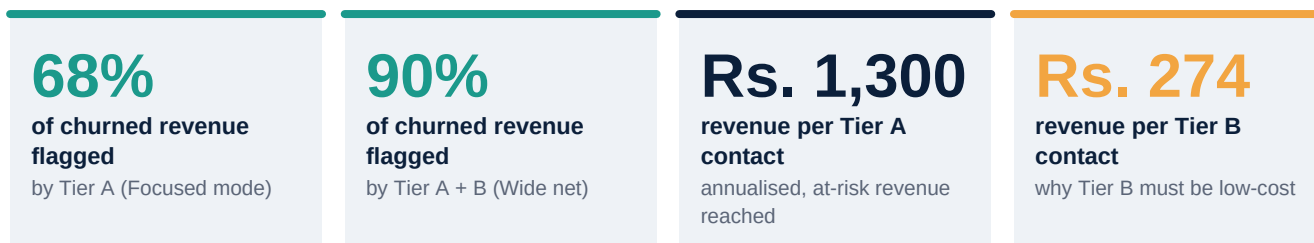
Recommended playbook

Tier	Action	Suggested treatment
A - Critical	Personal, same-day outreach	Route to a retention desk within 24 hours. Prioritise subscribers with an MNP enquiry; resolve open complaints and failed payments first; then make a tailored retention offer.
B - Watch	Automated, low-cost nudges	Pre-lapse recharge reminders (around day 45-60 of the recharge cycle), payment-retry assistance, autopay and bundle prompts. Escalate to Tier A if the score rises.
C - Stable	No retention spend	Business as usual. Keep monitoring; nothing to gain from discounting customers who are not at risk.

6. Business value

Where the model turns into revenue — and how to size the prize with a controlled pilot.

Subscribers who churned in the observed 90-day window represent about **Rs. 59.6 lakh** of annualised revenue (last-month ARPU x 12). Churners are, on average, lower-value than the base (Rs. 1,840 vs. Rs. 2,603 per year), which is why targeting matters: spending heavily to keep the wrong customers is easy to do and hard to justify.



Illustrative annual revenue retained

The value depends on how many flagged churners a retention offer actually saves. The table shows what different save rates would mean for the churn volume seen in one 90-day window. **These are scenarios, not forecasts** — the pilot below will replace them with a measured figure.

Save rate among flagged churners	Tier A only	Tier A + B (Wide net)
10%	Rs. 4.1 lakh	Rs. 5.4 lakh
20%	Rs. 8.1 lakh	Rs. 10.7 lakh
30%	Rs. 12.2 lakh	Rs. 16.1 lakh

Flagged annualised revenue at risk: Rs. 40.7 lakh (Tier A) and Rs. 53.6 lakh (Tier A + B). Assumes a saved subscriber keeps paying their current ARPU for 12 months.

A simple break-even rule for outreach cost

Each Tier A contact reaches about **Rs. 1,300** of annualised at-risk revenue. At a 20% save rate that is worth roughly **Rs. 260** per contact — so a retention call or offer costing less than that pays for itself.

Tier B contacts reach about Rs. 274 each (about Rs. 55 at a 20% save rate), which is why Tier B should stay automated and inexpensive.

Proposed pilot

- **Score** the full base monthly and split Tier A (and B) at random into a **treatment** group and a **hold-out** group that receives no special outreach.
- **Measure** the churn-rate difference and revenue retained over 90 days — the only reliable way to know the true save rate and payback.
- **Decide** the roll-out size, offer budget and tier cut-offs from the measured results.

7. Reliability, assumptions and next steps

We tested whether the results are real before recommending them to you.

Check	Result	Why it matters
Unseen test data	ROC-AUC 0.977; train 0.973, validation 0.971	Near-identical scores on all three groups: the model is not memorising the training data.
5-fold cross-validation	ROC-AUC 0.970 (+/- 0.003); PR-AUC 0.814	Stable across different slices of the data — not a lucky split.
Shuffled-label test	ROC-AUC 0.497 (5 runs)	With the churn labels scrambled, the model falls to coin-toss level. The real result comes from real patterns.
Trivial baselines	'Nobody leaves': recall 0%. Random guess: ROC-AUC 0.51	Our model's F1 is 0.73 against 0.00 and 0.07 for these baselines.
Leakage and overlap	No outcome fields used; no subscriber in more than one split; strongest single field scores 0.81	No single field gives the answer away.
Drop the top two signals	ROC-AUC still 0.936	Predictive power is spread across many real behaviours, not two fields.

Assumptions and limitations

- **This is a baseline.** Logistic regression was chosen for transparency. More advanced models (for example gradient boosting) are likely to add further lift, and we will benchmark them against these results.
- **Timing of the inputs.** Results come from one historical snapshot. Before go-live we will confirm with the data owners that signals such as recharge recency, MNP enquiries and offer flags are captured *before* the churn window they are meant to predict, so live scores behave like these results. The pilot doubles as this confirmation.
- **Financial figures are scenarios.** Revenue is annualised last-month ARPU; save rates depend on the offers and channels used and must be measured in the pilot.
- **Churn drivers describe association, not proof of cause.** A hold-out group in the pilot is what proves that outreach changes the outcome.

Next steps

When	What	Outcome
Weeks 1-2	Confirm data timing and the scoring feed; agree tier cut-offs and outreach capacity.	Go-live checklist signed off
Weeks 3-6	Run the Tier A / Tier B pilot with a hold-out group.	Measured save rate and payback
Weeks 7-10	Benchmark advanced models; calibrate scores; add explanation of each subscriber's top risk factors for agents.	Higher lift, agent-ready reasons
Ongoing	Monthly re-scoring, drift monitoring, quarterly refresh.	Model stays accurate as behaviour shifts

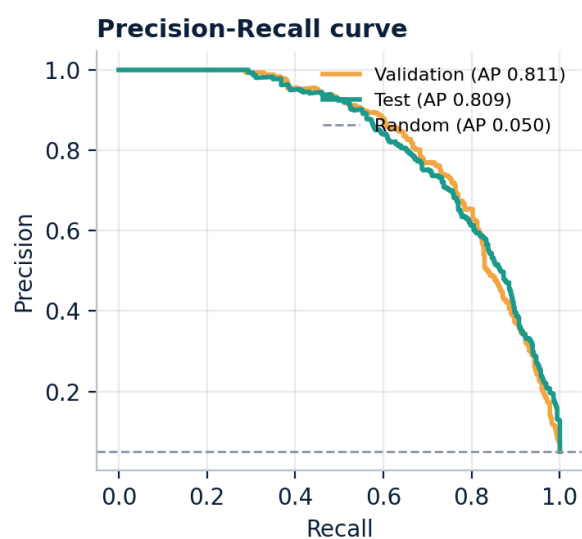
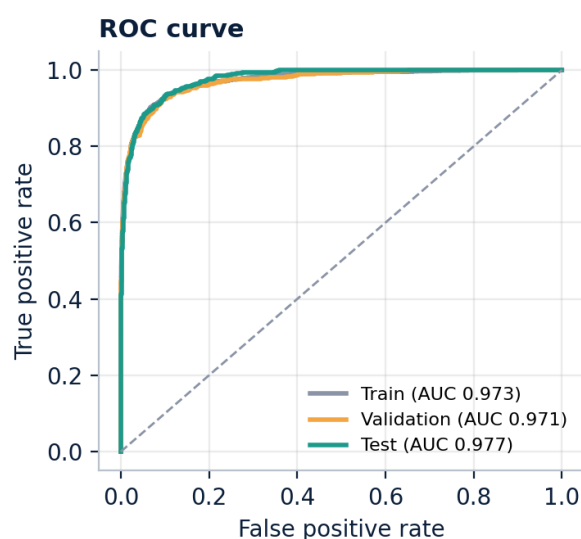
Appendix: technical summary

For the client's analytics and data teams.

Item	Detail
Target	churn_flag_90d (1 = churned within 90 days); positive rate 5.01%
Excluded columns	churn_reason, churn_date, churn_flag_30d (outcome-linked); subscriber_id, join_date (captured by tenure), circle_code (duplicate of circle)
Features	36: 31 numeric / binary (median-imputed, standardised) and 5 categorical (circle, zone, plan_type, device_brand, home_product; one-hot encoded)
Split	Stratified 70 / 15 / 15: 45,316 train / 9,711 validation / 9,711 test; random seed 42
Model	Logistic regression, L2 regularisation, C = 0.1 (chosen on validation ROC-AUC from 0.01, 0.1, 1, 10), class_weight = balanced, lbfgs solver
Decision cut-off	Tuned on validation to maximise F1: 0.9151. Default 0.50 also reported.
Tools	Python, pandas, scikit-learn, matplotlib. Fully reproducible from the delivered scripts.

All metrics by split

Split	Cut-off	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Train	0.50	0.9251	0.3927	0.9066	0.5480	0.9725	0.8196
Train	0.92	0.9758	0.7884	0.7060	0.7449	0.9725	0.8196
Validation	0.50	0.9193	0.3726	0.8971	0.5266	0.9711	0.8111
Validation	0.92	0.9750	0.7624	0.7263	0.7439	0.9711	0.8111
Test	0.50	0.9220	0.3825	0.9074	0.5381	0.9769	0.8090
Test	0.92	0.9734	0.7478	0.7078	0.7273	0.9769	0.8090



ROC and precision-recall curves for the final model. Performance is consistent across train, validation and test.